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Introduction to Data Science

Genre-Based User Clustering
for Movie Recommendation

Group 8

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Abstract

This project investigates genre-based user segmentation using the MovieLens small dataset, which contains 100,836 ratings from 610 users. Movie ratings are aggregated into a 19-dimensional user-genre representation. Mean-centering expresses each genre score relative to the user's overall rating level, and the resulting features are standardized before clustering.

K-Means is used as the main clustering method. The number of clusters is selected by examining SSE, Silhouette Score, cluster sizes, and variation across random seeds. Based on these criteria, $K = 5$ is adopted as a practical segmentation choice. Agglomerative Clustering with Ward linkage is then evaluated on the same standardized data with the same number of clusters. The final K-Means model achieves a Silhouette Score of 0.0638, compared with 0.0140 for Ward. Although K-Means performs better under this criterion in the tested setting, both scores indicate weak separation.

The cluster-average profiles reveal differences in genre preferences, but these averages do not fully describe individual members. A cluster-based recommendation module illustrates how the segmentation can be used to generate movie suggestions. Recommendation quality is not evaluated on held-out ratings, so the results support conclusions about the observed segmentation rather than recommendation accuracy.

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1 Introduction

Recommendation systems use information about users and items to generate suggestions from a larger catalogue. In a movie-rating dataset, one possible source of information is the rating behaviour of other users. However, comparing users directly at the movie level is difficult when their rating histories contain relatively few shared movies.

This project examines whether genre-level rating profiles can provide an interpretable basis for user segmentation. Each user is represented by 19 mean-centered genre scores, and K-Means is applied to the standardized profiles. The resulting five-cluster solution is compared with Agglomerative Ward under the same input representation and cluster count.

The main analysis task is user clustering. The recommendation module is a downstream demonstration: it uses ratings from other members of the assigned cluster to generate suggestions. The experiments evaluate the clustering results, but do not establish whether this recommendation strategy outperforms a non-personalized baseline.

1.1 About Dataset

The MovieLens small dataset [1] contains 100,836 ratings from 610 users across 9,742 movies. Each movie is labeled with one or more genres, such as Action, Comedy, Drama, Romance, Sci-Fi, Thriller, and others. The main files used in this project are:

- `movies.csv`: contains movie identifiers, titles, and genres.
- `ratings.csv`: contains user identifiers, movie identifiers, rating values, and timestamps.
- `tags.csv`: contains user-generated tags.
- `links.csv`: contains external movie identifiers.

This project uses only `movies.csv` and `ratings.csv`, since the goal is to learn user genre preferences from rating behavior. Tags are not part of the final feature matrix.

1.2 Motivation

The original user–movie representation has 9,742 possible movie features per user, whereas only 100,836 ratings are observed across all 610 users. Genre aggregation provides a smaller representation by combining ratings for movies that share genre labels. Each user can therefore be described using 19 features rather than thousands of movie-specific entries.

Users also differ in their overall rating levels. Mean-centering reduces the influence of these differences by expressing a genre’s average rating relative to the user’s own average. This representation describes relative rating patterns; it does not measure preference independently of the movies that the user has chosen to rate.

Clustering is used to summarize these profiles into a small number of groups. The intended benefit is an interpretable description of broad preference patterns and a possible source of within-group ratings for recommendation. Whether this grouping improves recommendation quality requires a separate evaluation.

1.3 Contribution

The main contributions of this project are:

- We construct a compact user–genre representation by aggregating movie ratings and expressing genre scores relative to each user’s overall rating level.
- We examine K-Means solutions for $K = 2, \dots, 14$ and select $K = 5$ by considering clustering metrics, cluster sizes, and variation across random seeds.

- We compare the final K-Means solution with Agglomerative Ward on the same standardized profiles at $K = 5$.
- We interpret the cluster-average genre profiles and discuss the distinction between differences in group averages and separation of individual users.
- We implement a cluster-based recommendation module and demonstrate its output, while identifying the additional evaluation needed to assess recommendation quality.

2 Related Work

2.1 Recommendation Research

Park et al. [2] reviewed 210 articles on recommender systems and classified them by publication year, journal, application field, and data-mining technique. Their review provides background for the recommendation setting of this project. However, it does not validate the particular user-genre representation or ranking rule used here.

2.2 Clustering Methods

K-Means represents each cluster by its mean vector and iteratively updates assignments and centroids to reduce the within-cluster sum of squares [4]. It provides a direct way to summarize numerical user profiles, although the resulting partition depends on the input representation, cluster count, and initialization.

Ward’s hierarchical method provides a different construction of the partition [3]. It begins with individual observations and successively merges groups according to a within-cluster dispersion criterion. Unlike K-Means, it does not iteratively reassign users between a fixed set of centroids.

2.3 Position of This Project

This project uses K-Means as the main model and Agglomerative Ward as a classical clustering baseline. Both methods receive the same standardized user-genre matrix and are compared at $K = 5$. Fixing the input and cluster count makes it possible to examine how the two algorithms partition the same profiles.

The experiment is a comparison of two established methods in a specific setting. It is not a benchmark of current state-of-the-art user segmentation methods. Recommendation is included as an application of the K-Means partition rather than as a separately validated prediction model.

3 Project Process

The main pipeline constructs user profiles, fits K-Means, and uses the resulting partition for recommendation. A separate comparison branch applies Agglomerative Ward to the same standardized profiles at $K = 5$ (Figure 1).

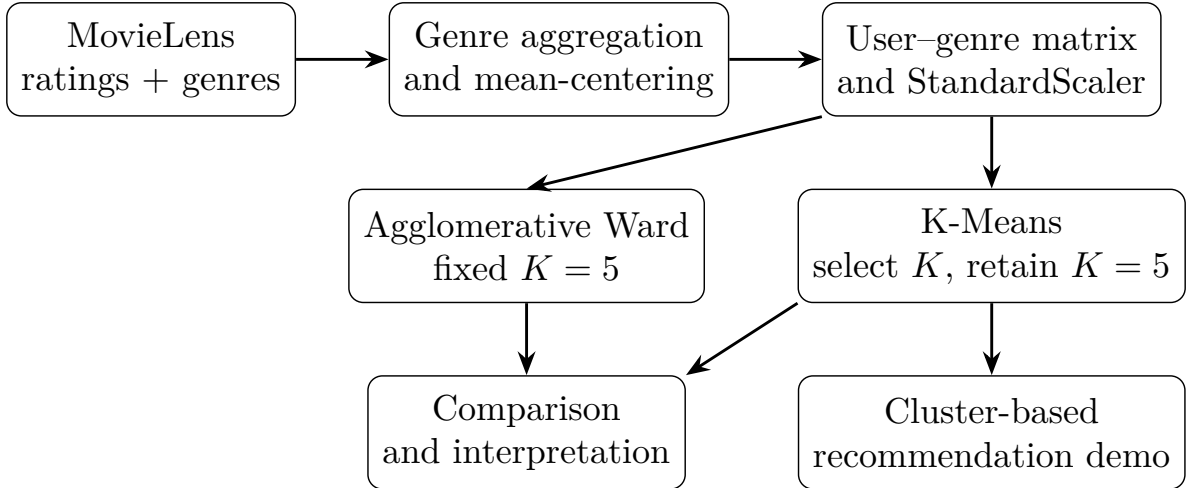


Figure 1: Main K-Means pipeline and the fixed- K Ward comparison. Recommendation uses the K-Means partition.

3.1 Data Preprocessing

The system uses two main CSV files from the MovieLens small dataset: `movies.csv` and `ratings.csv`. The movie file contains 9,742 movies with three main columns: `movieId`, `title`, and `genres`. The rating file contains 100,836 ratings from 610 users on a scale from 0.5 to 5.0.

Table 1: MovieLens small dataset statistics

Metric	Value
Total movies	9,742
Total users	610
Total ratings	100,836
Rating scale	0.5–5.0
Number of genres	19
Average ratings per user	≈ 165
Movies rated per user	min 20, max 2,698

The `genres` attribute in the MovieLens dataset is stored as a pipe-separated string, such as:

Adventure|Animation|Children|Comedy|Fantasy

To convert this categorical information into numerical features, the genre string is transformed into binary indicator variables using one-hot encoding. Each genre becomes a separate column, where a value of 1 indicates that the movie belongs to that genre and 0 otherwise.

Table 2: Example of one-hot genre encoding

Title	Action	Adventure	Animation	Comedy	Fantasy
Toy Story (1995)	0	1	1	1	1
Jumanji (1995)	0	1	0	0	1
Heat (1995)	1	0	0	0	0

After one-hot encoding, the genre indicators are merged with the rating table using `movieId`. The resulting data contains one row for each user-movie rating event, together with the rating value and the corresponding movie genre indicators.

3.2 Centered Rating

Users may differ in their overall rating levels: some give relatively high scores across their rated movies, while others give lower scores. Using raw genre averages can therefore mix genre-specific differences with differences in users’ overall rating levels.

To reduce this influence, the mean rating for each user–genre pair is expressed relative to the user’s overall mean:

$$\tilde{r}_{u,g} = \bar{r}_{u,g} - \bar{r}_u \quad (1)$$

where $\bar{r}_{u,g}$ is the mean raw rating that user u gives to movies belonging to genre g , and $\bar{r}_u$ is the overall mean rating of user u across all rated movies.

Formally, let $M_{u,g}$ be the set of movies rated by user u that belong to genre g , and let M_u be the set of all movies rated by user u . Then:

$$\bar{r}_{u,g} = \frac{1}{|M_{u,g}|} \sum_{m \in M_{u,g}} r_{u,m}, \quad \bar{r}_u = \frac{1}{|M_u|} \sum_{m \in M_u} r_{u,m} \quad (2)$$

When user u has not rated any movie belonging to genre g , $\bar{r}_{u,g}$ is undefined. The implementation fills the corresponding centered entry with zero so that a complete numerical matrix can be passed to the clustering algorithms. This is an imputation assumption, not an observation that the user has a neutral preference.

For an observed user–genre pair, a positive value means that the genre’s average rating is above the user’s overall average, while a negative value means that it is below that average. These scores describe relative rating patterns and should not be interpreted as direct measures of liking or disliking every movie in a genre.

Because an unobserved entry is also filled with zero, a zero-valued feature can represent either an observed genre average equal to the user’s mean or a lack of genre-specific rating evidence.

Table 3: Centered rating examples for selected users and genres

User	$\bar{r}_u$	Action	Animation	Comedy	Drama	Romance	Thriller
1	4.37	−0.044	0.323	−0.089	0.163	−0.059	−0.221
2	3.95	0.006	0.000	0.052	−0.066	0.552	−0.248
3	2.44	1.136	−1.936	−1.436	−1.686	−1.936	1.707
4	3.56	−0.236	0.444	−0.046	−0.072	−0.176	−0.003
5	3.64	−0.525	0.697	−0.170	0.164	−0.545	−0.081

3.3 User-Genre Matrix

Using the centered genre preferences, we construct a User-Genre matrix:

$$R \in \mathbb{R}^{N \times G} \quad (3)$$

where $N = 610$ is the number of users and $G = 19$ is the number of genres. Each entry $R_{u,g}$ represents the centered preference of user u for genre g . Each row of the matrix is a 19-dimensional preference vector:

$$R_u = [R_{u,1}, R_{u,2}, \dots, R_{u,19}] \quad (4)$$

This vector summarizes the observed genre-level rating pattern of a user, with unobserved entries filled as described above. Compared with the original User-Movie matrix, which has 9,742 movie dimensions, the User-Genre matrix is much smaller and easier to interpret.

Table 4: Properties of the User-Genre matrix

Property	Value
Rows	610 users
Columns	19 genres
Value range	$[-3.54, 2.56]$
Fill value for unobserved user-genre entries	0.0
Active-user threshold	10 ratings
Active users retained	610

Users with fewer than 10 total ratings are considered weak-profile users because their genre vectors may be too sparse. However, in this dataset, all 610 users have at least 20 ratings, so all users are retained for clustering.

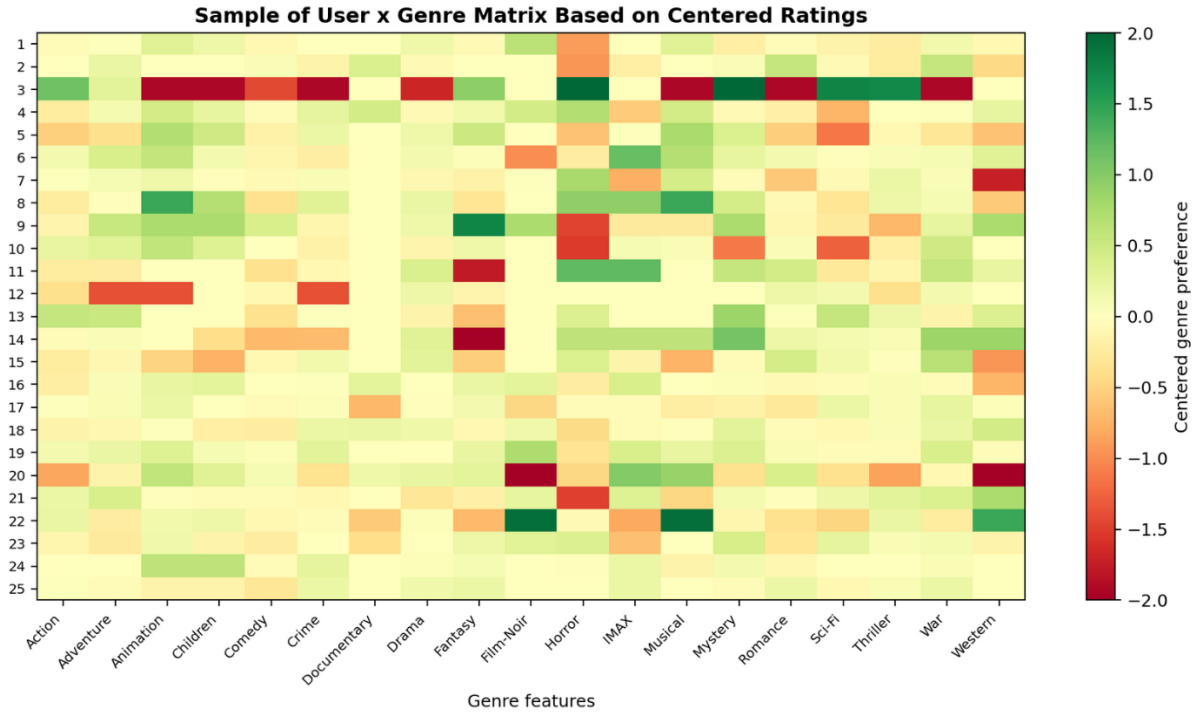


Figure 2: Sample of the User-Genre matrix based on centered ratings

3.4 Feature Scaling

Before clustering, the User-Genre matrix is standardized using z-score normalization. Both K-Means and Ward are applied to this same scaled representation. Since feature variance affects Euclidean distances, standardization places the genre columns on comparable variance scales. For each genre feature g , the standardized value is computed as:

$$z_{u,g} = \frac{R_{u,g} - \mu_g}{\sigma_g} \quad (5)$$

where μ_g and σ_g are the mean and standard deviation of genre g computed over all users. After scaling, each genre column has approximately zero mean and unit variance. The scaled matrix is denoted as:

$$Z \in \mathbb{R}^{610 \times 19} \quad (6)$$

The matrix Z is used for fitting and evaluating both clustering methods. For interpretation, cluster averages are reported in the original centered-rating units rather than standardized units.

Table 5: Effect of StandardScaler on selected genre columns

Genre	Before Scaling		After Scaling	
	μ_g	σ_g	μ_g	σ_g
Action	-0.102	0.345	0.000	1.000
Animation	-0.004	0.600	0.000	1.000
Documentary	0.062	0.431	0.000	1.000
Film-Noir	0.089	0.480	0.000	1.000

3.5 K-Means Clustering

K-Means partitions the 610 users into K clusters by minimizing the Within-Cluster Sum of Squares:

$$\text{WCSS}(K) = \sum_{k=1}^K \sum_{u: y_u=k} \|z_u - c_k\|^2 \quad (7)$$

where z_u is the scaled genre preference vector of user u , c_k is the centroid of cluster k , and y_u is the cluster assignment of user u . The centroid of each cluster is computed as the mean of all users assigned to that cluster:

$$c_k = \frac{1}{|C_k|} \sum_{u: y_u=k} z_u \quad (8)$$

The K-Means algorithm follows two main steps:

1. **Assignment step:** each user is assigned to the nearest centroid:

$$y_u = \arg \min_k \|z_u - c_k\|^2 \quad (9)$$

2. **Update step:** each centroid is updated as the mean vector of all users assigned to that cluster:

$$c_k = \frac{1}{|C_k|} \sum_{u: y_u=k} z_u \quad (10)$$

These steps are repeated until the stopping condition is reached. The implementation uses `sklearn.cluster.KMeans` with `n_init = 10`. The selected final model uses `random_state = 42`.

Candidate values $K = 2, \dots, 14$ are examined using SSE, Silhouette Score, and cluster sizes. Additional runs with different random seeds are used to examine variation in the metrics and minimum cluster size for $K = 5, \dots, 8$. The selection procedure and its results are reported in Section 5.2.

The final model uses $K = 5$. This is a practical choice within the exploratory analysis, rather than the value that maximizes Silhouette Score over all tested cluster counts.

3.6 Agglomerative Clustering with Ward Linkage

Agglomerative Ward is used as the comparison method. It starts with each user in a separate cluster and repeatedly merges the pair whose union produces the smallest increase in within-cluster sum of squares.

For clusters A and B , the increase is

$$\Delta(A, B) = \frac{|A||B|}{|A| + |B|} \|\mu_A - \mu_B\|^2, \quad (11)$$

where μ_A and μ_B are the mean standardized profiles of the two clusters. This criterion is applied to the same standardized matrix Z used by K-Means.

The implementation uses `sklearn.cluster.AgglomerativeClustering` with `n_clusters = 5` and `linkage = 'ward'`. Ward has no random centroid initialization. Once a merge is made, the hierarchy does not revise that merge by reassigning individual users.

The cluster count is fixed at five to match the selected K-Means model. Ward is used for comparison only; its partition is not used by the recommendation module.

3.7 Cluster-Based Movie Recommendation

After clustering, each user is assigned to a group of users with similar genre-preference patterns. The recommendation module is built on top of these clusters. For an existing active user u assigned to cluster C_k , candidate movies are obtained from ratings provided by other users in the same cluster.

Candidate generation and filtering. For each candidate movie m , we compute the number of ratings contributed by other users in the target cluster:

$$\text{rating_count}(m) = |\{v \in C_k \setminus \{u\} : v \text{ has rated } m\}|. \quad (12)$$

The corresponding average rating is

$$\text{avg_rating}(m) = \frac{1}{\text{rating_count}(m)} \sum_{\substack{v \in C_k \setminus \{u\} \\ v \text{ rated } m}} r_{v,m}. \quad (13)$$

Let $W(u)$ denote the set of movies already watched by user u . Movies already watched by the target user are removed. To avoid recommendations based on very weak evidence, a candidate must also receive at least τ ratings from users in the cluster, where the current implementation uses $\tau = 3$. The final candidate set is therefore

$$M_{\text{cand}} = \{m : \text{rating_count}(m) \geq \tau\} \setminus W(u). \quad (14)$$

Weighted scoring and Top- N ranking. Each remaining candidate is ranked using

$$\text{score}(m) = \text{avg_rating}(m) \log(1 + \text{rating_count}(m)). \quad (15)$$

The score combines average rating with the number of supporting ratings within the cluster. For a fixed average rating, additional ratings increase the score, but their contribution grows logarithmically rather than linearly. The candidates are sorted by this score in descending order, and the top- N movies are returned.

This is a heuristic ranking rule. The rating-count term is not a statistical confidence estimate, and the experiment does not establish that this rule is better than ranking by average rating or popularity.

Table 6: Recommendation strategy for different user cases

User case	Recommendation strategy
Existing active user	Use the assigned K-Means cluster, exclude watched movies, and rank candidate movies using the weighted score.
New user with favorite genres	Construct an approximate preference vector from the selected genres, standardize it, assign it to the nearest K-Means centroid, and recommend movies from that cluster.
New user without favorite genres	Use the popularity-based fallback because no preference information is available.
Weak-profile user	Use the popularity-based fallback because the available rating history is insufficient for a reliable genre-preference profile.

User cases. The recommendation strategy depends on the amount of information available for the user, as summarized in Table 6.

For a new user who selects favorite genres, the implementation assigns a value of 1 to the selected genres and 0 to the others, then applies the fitted scaler and finds the nearest K-Means centroid. These inputs are not derived from observed ratings in the same way as the training profiles. The procedure is therefore an approximate onboarding rule, whose effectiveness has not been evaluated.

4 Dataset Analysis

4.1 Rating Distribution

Figure 3 shows the distribution of rating values in the dataset. Most ratings are concentrated around 3.0, 3.5, and 4.0, indicating that users tend to give medium to high ratings more frequently than very low ratings.

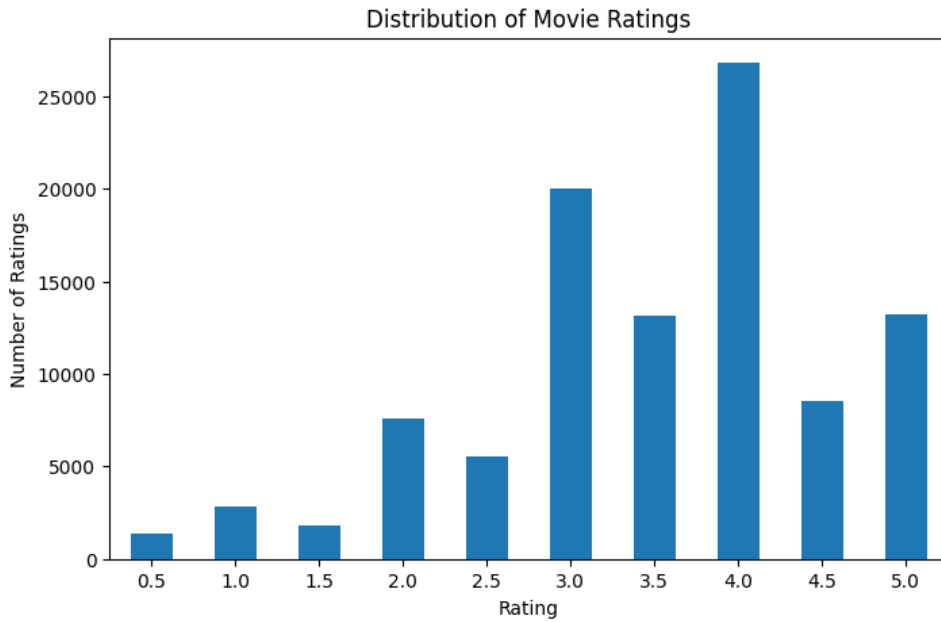


Figure 3: Rating Distribution

4.2 User Rating Behavior

Figure 4 shows an uneven distribution of rating counts. Users have between 20 and 2,698 ratings, so the amount of evidence available for constructing their profiles differs considerably.

Genre aggregation reduces the dimensionality of the representation, but it does not make all user profiles equally well supported. A genre average based on one rating may be less informative than an average based on many ratings. Mean-centering addresses differences in overall rating levels; it does not correct unequal rating counts.

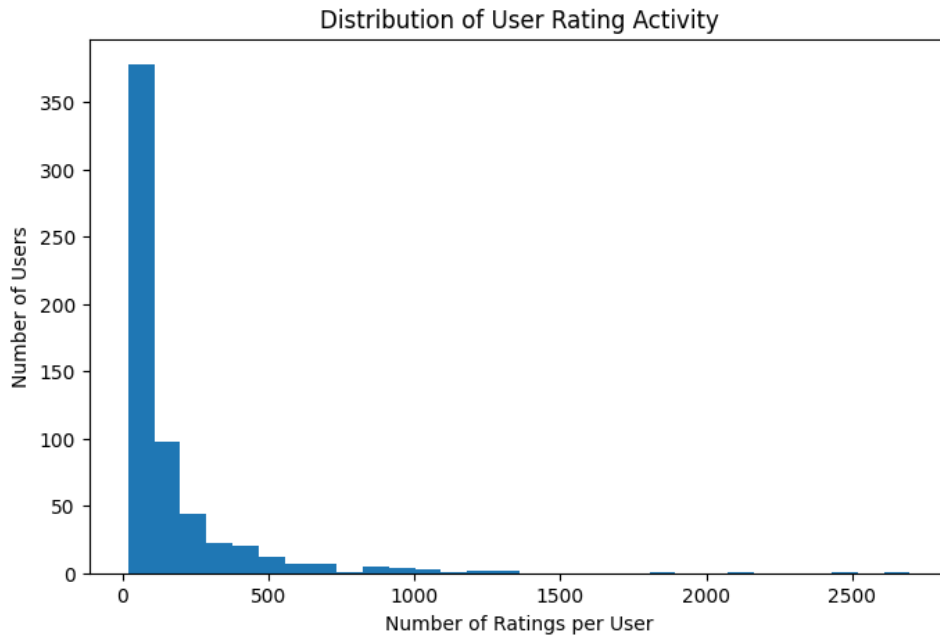


Figure 4: Distribution of the number of ratings per user

4.3 Genre Distribution

Figure 5 shows that genre labels are unevenly represented in the movie catalogue. Drama and Comedy occur much more frequently than genres such as Film-Noir, IMAX, and Western. Since movies can have multiple labels, a movie may contribute to more than one genre count.

This imbalance matters for the amount of evidence available for user-genre averages. Standardization makes feature variances comparable, but does not equalize the number of ratings supporting each feature. The present experiments do not isolate the effect of genre imbalance on clustering or recommendation quality.

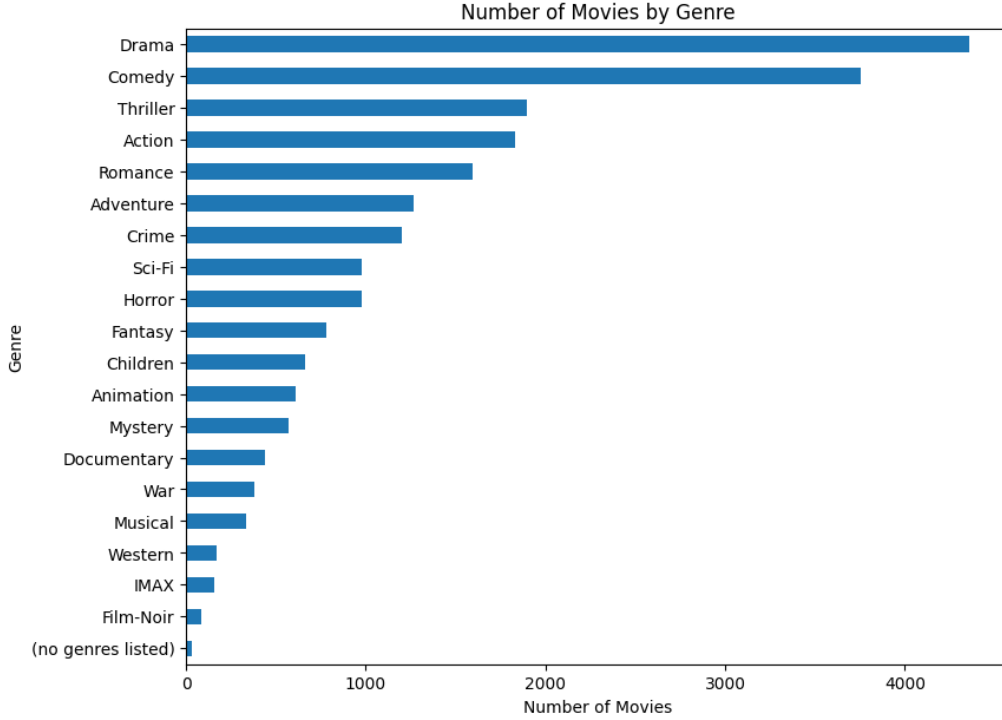


Figure 5: Genre Distribution

5 Experiments

5.1 Experimental Protocol

The experiments examine the K-Means cluster count, compare the selected partition with Ward, and interpret the resulting user profiles. Both methods are evaluated on the complete standardized user–genre matrix. No held-out recommendation test is conducted.

1. **Profile construction:** Ratings and movie genres are transformed into a 610×19 mean-centered user–genre matrix. Unobserved user–genre entries are filled with zero. All users satisfy the threshold of 10 ratings.
2. **Feature scaling:** The genre columns are standardized using `StandardScaler`. The same matrix is used by both clustering algorithms.
3. **Initial K-Means evaluation:** Values $K = 2, \dots, 14$ are evaluated with `random_state = 42` and `n_init = 10`, using SSE, Silhouette Score, and cluster sizes.
4. **Variation across initializations:** Candidate values $K = 5, \dots, 8$ are evaluated with random seeds 0 through 9 and `n_init = 10`. The mean and standard deviation of Silhouette Score and minimum cluster-size statistics are recorded.
5. **Selection of the final model:** $K = 5$ is retained based on the exploratory criteria above. The final K-Means partition uses `random_state = 42`.
6. **Fixed- K comparison:** Agglomerative Ward is fitted to the same input with $K = 5$. The two partitions are compared using Euclidean Silhouette Score, cluster sizes, and mean centered genre profiles.

7. **Interpretation and demonstration:** Cluster averages are examined in centered-rating units. A recommendation example illustrates how the final K-Means partition supplies candidate ratings.

The clustering metrics describe the partitions of the observed user profiles. They do not measure performance on unseen ratings or establish the accuracy of the recommendation module.

5.2 Selection of K

For each value of K from 2 to 14, K-Means is trained with `random_state = 42` and `n_init = 10`. The number of clusters is evaluated using the Sum of Squared Errors (SSE), Silhouette Score, cluster size, and stability across multiple random seeds.

Sum of Squared Errors (SSE). SSE measures the compactness of the clusters:

$$\text{SSE} = \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - w_k\|^2, \quad (16)$$

where C_k denotes cluster k and w_k is its centroid. A lower SSE means that users are, on average, closer to their assigned cluster centroids. However, SSE naturally decreases as K increases. Therefore, the Elbow Method is used to identify a reasonable range of K rather than to determine the final value by itself.

Silhouette Score. The Silhouette Score measures both within-cluster cohesion and between-cluster separation. For user i , it is defined as

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \quad (17)$$

where $a(i)$ is the mean distance from user i to other users in the same cluster, and $b(i)$ is the smallest mean distance from user i to users in any other cluster. Higher values indicate better cohesion and separation.

Figure 6 shows the SSE and Silhouette Score for $K = 2, \dots, 14$. The SSE decreases gradually as K increases and does not show a single sharp elbow. Therefore, it is considered together with the Silhouette Score and cluster-size analysis rather than being used as the sole selection criterion.

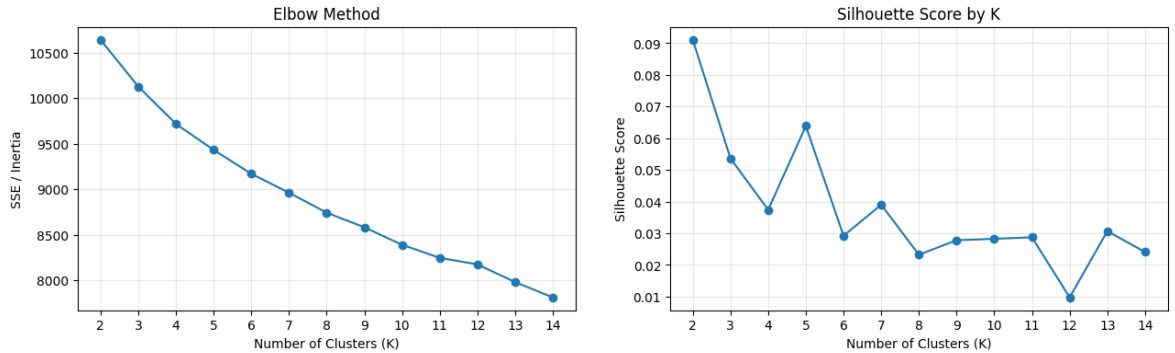


Figure 6: SSE (Elbow Method) and Silhouette Score for $K = 2, \dots, 14$.

In addition to these clustering metrics, cluster sizes are inspected because the clusters are subsequently used by the recommendation module. Very small clusters restrict the number of

other users who can contribute ratings for cluster-based movie recommendation; this may limit candidate support. The stability of candidate values of K is therefore also evaluated across multiple random seeds.

Table 7: K-Means results for different values of K (single run, `random_state` = 42)

K	SSE	Min cluster size	Max cluster size
2	10,639.9	228	382
3	10,126.9	133	270
4	9,716.5	55	255
5	9,430.9	25	279
6	9,168.6	17	229
7	8,961.1	24	200
8	8,741.3	1	193
10	8,389.0	1	172
12	8,173.8	1	168
14	7,811.4	1	173

The single-run results show that increasing K continuously reduces SSE, as expected. However, larger values of K also tend to produce increasingly small clusters. In particular, $K = 8, 10, 12$, and 14 already produce singleton clusters in the single run. This fragmentation is undesirable for the downstream recommendation task because recommendations are generated using ratings from other users in the same cluster.

Based on the initial analysis, moderate values $K = 5, \dots, 8$ are examined further using multiple random seeds. Table 8 summarizes the mean and standard deviation of the Silhouette Score together with the minimum cluster-size statistics across these runs. The ten seeds are 0 through 9. These statistics describe variation in metric values and cluster sizes, not agreement between user assignments.

Table 8: Multi-seed stability results for candidate values of K

K	Silhouette Mean	Silhouette STD	Avg. Min Cluster Size	Worst Min Cluster Size
5	0.0429	0.0139	32.8	18
6	0.0264	0.0052	29.3	16
7	0.0288	0.0053	20.2	1
8	0.0262	0.0065	12.2	1

Among the candidates $K = 5, \dots, 8$, $K = 5$ has the highest mean Silhouette Score, at 0.0429 with a standard deviation of 0.0139. Its minimum cluster size ranges down to 18 users across the tested seeds, compared with singleton clusters observed for $K = 7$ and $K = 8$. The selected final run produces cluster sizes of 107, 279, 63, 25, and 136 users.

These results support choosing $K = 5$ over the larger candidates examined in the multi-seed analysis. They do not show that five clusters are optimal across all possible selection criteria. In particular, $K = 2$ has the highest Silhouette Score in the initial single-seed analysis.

Five clusters are retained to examine more detailed preference profiles while avoiding the singleton groups observed for some larger values of K . This choice reflects the intended level of segmentation. The experiment does not demonstrate that five groups provide better recommendations than two groups.

The multi-seed results describe variation in Silhouette Score and minimum cluster size. No partition-agreement measure is computed, so they should not be interpreted as evidence that

individual users retain the same assignments across seeds. The selected value $K = 5$ is therefore reported as a practical choice within this exploratory analysis.

5.3 Cluster Interpretation

Figure 7 presents the mean centered genre ratings of users in each final K-Means cluster. The model is fitted in standardized space, while these profiles are shown in the original centered-rating units for interpretation.

Table 9: Final K-Means cluster sizes and the five genres with the highest mean centered ratings within each cluster

Cluster	Users	Highest-ranked genres
0	107	Crime, Film-Noir, Drama, Mystery, War
1	279	War, Animation, IMAX, Film-Noir, Musical
2	63	Fantasy, Animation, Children, Adventure, IMAX
3	25	War, Drama, Mystery, Film-Noir, Documentary
4	136	Mystery, Thriller, IMAX, Crime, Sci-Fi
Total	610	

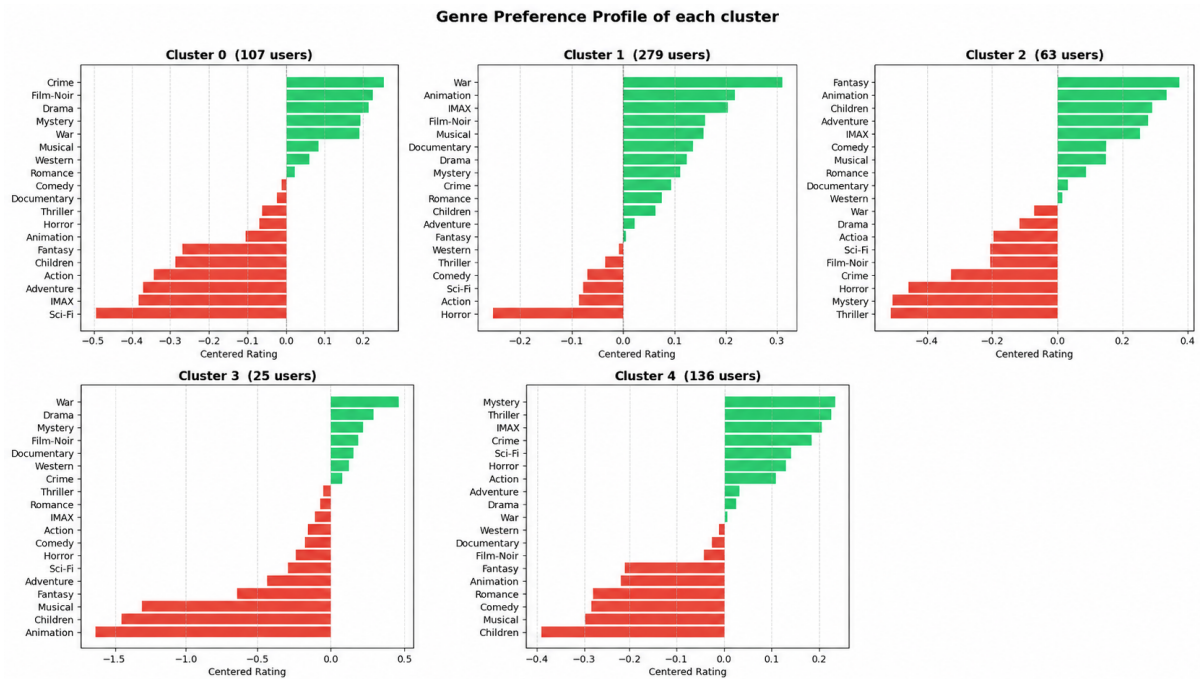


Figure 7: Mean centered genre profiles of the final K-Means clusters. Positive and negative values indicate ratings above and below users' individual average levels, respectively.

The profiles show several differences in group averages. Cluster 0 has positive mean scores for Crime, Film-Noir, and Drama, alongside negative scores for Sci-Fi and Adventure. Cluster 1, the largest group, has positive scores for War, Animation, and IMAX, with relatively modest deviations for many other genres.

The contrast between Clusters 2 and 4 is particularly visible. Cluster 2 has positive mean scores for Fantasy, Animation, Children, and Adventure, but negative scores for Mystery and Thriller. Cluster 4 shows positive scores for Mystery and Thriller, while Children and Animation have negative scores.

Cluster 3 is distinguished by strongly negative mean centered ratings for Animation, Children, and Musical. Its War and Drama scores are positive. This profile differs from some other groups in the magnitude of its negative scores as well as in the ordering of genres.

These observations concern cluster averages. They do not establish that each member follows the average pattern or that the groups are clearly separated. The low Silhouette Score reported in Section 5.4 provides evidence that the partition has weak cohesion and separation under the evaluated distance criterion.

The profiles also include zero-filled unobserved entries. Consequently, a mean close to zero may combine observed relative ratings with missing genre evidence. These averages should be read as summaries of the constructed features rather than complete descriptions of users' preferences.

5.4 Comparison with Agglomerative Ward

After selecting the K-Means model, Agglomerative Ward is evaluated with the same standardized input and $K = 5$. The comparison uses the final K-Means run with `random_state = 42` and `n_init = 10`. Ward uses `linkage = 'ward'` and does not require random centroid initialization.

Table 10 reports the results. The K-Means Silhouette Score of 0.0638 is from the selected final run. It differs from the mean value of 0.0429 in Table 8, which summarizes ten different seeds.

Table 10: Comparison on the same standardized profiles at $K = 5$. Cluster sizes are measured in users.

Method	Silhouette	Min size	Max size	Singletons
K-Means	0.0638	25	279	0
Agglomerative Ward	0.0140	10	235	0

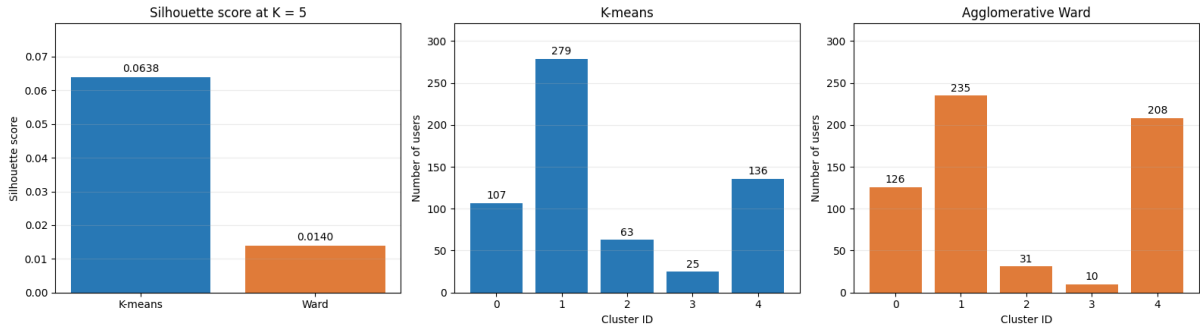


Figure 8: Silhouette Scores and cluster sizes at $K = 5$. Cluster identifiers are specific to each method and do not identify corresponding groups across the two partitions.

K-Means achieves a higher Silhouette Score than Ward in this experiment. Under the Euclidean Silhouette criterion, its partition therefore has better average cohesion and separation. However, both scores are close to zero, so the comparison does not indicate strong separation for either method.

The smallest K-Means cluster contains 25 users, compared with 10 for Ward. A larger group offers more potential contributors of ratings, but cluster size alone does not establish the number or quality of eligible movie candidates. K-Means also has a larger maximum cluster, so the results do not support a general claim that all its groups are more balanced.

Figure 9 shows that both methods include users with near-zero and negative Silhouette values. Negative values indicate that a user's average distance to another cluster is smaller than

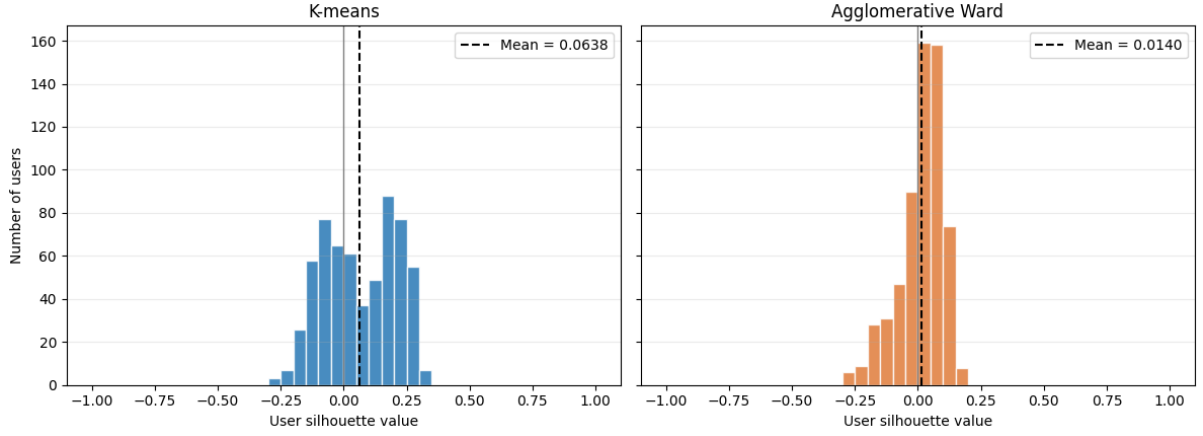


Figure 9: Distributions of user Silhouette values. Dashed lines mark the method-specific means.

its average distance to the assigned cluster. The distributions reinforce the limitation visible in the low overall scores.

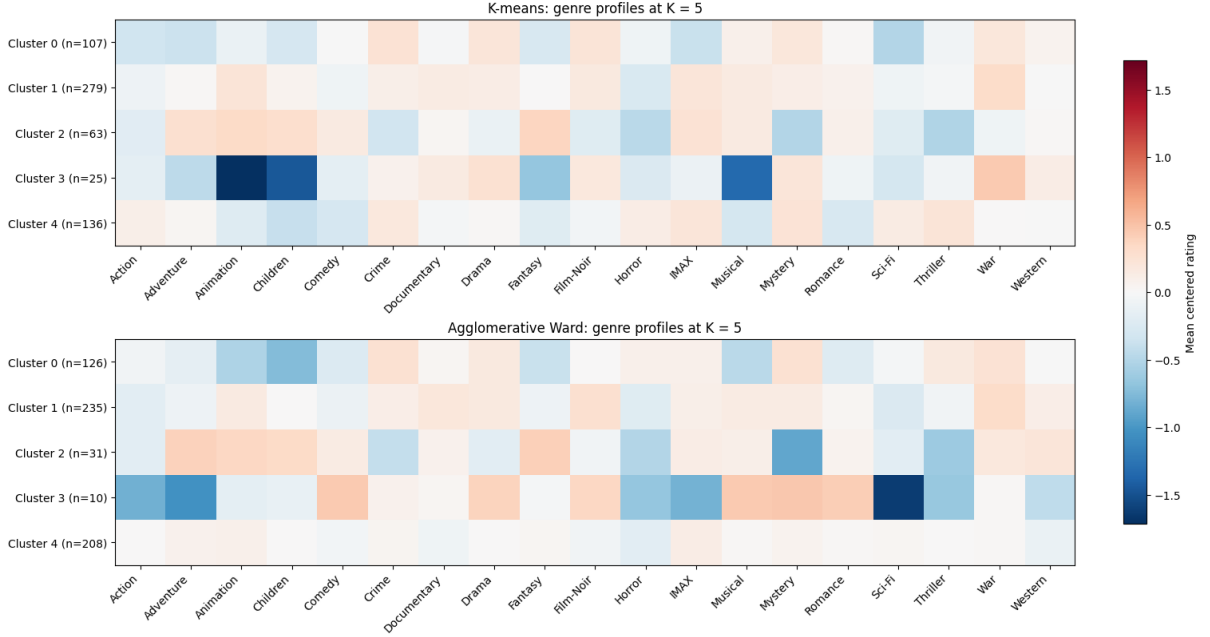


Figure 10: Cluster-average genre profiles at $K = 5$. Both panels use the same color scale. Cluster identifiers are not matched between methods.

The heatmaps show that the algorithms produce different group summaries. For example, both partitions contain a group with positive Fantasy, Animation, and Children scores. Ward also produces a ten-user group with a strongly negative Sci-Fi average. Similarities in average profiles do not establish that the corresponding groups contain the same users.

Taken together, the higher Silhouette Score and larger minimum cluster size support retaining K-Means within this five-cluster setting. The comparison does not identify the best cluster count for Ward, establish superiority across other representations, or demonstrate better recommendation accuracy.

5.5 Recommendation Demonstration

To demonstrate the downstream application of the clustering result, an existing active user is selected as an example. User 50 has rated 310 movies, has an average rating of 2.78, and is assigned by K-Means to Cluster 0.

Following the recommendation procedure described in Section 3.7, movies already watched by the user are excluded. Candidate movies rated by other users in Cluster 0 are then ranked using the weighted score

$$\text{score}(m) = \text{avg_rating}(m) \log(1 + \text{rating_count}(m)). \quad (18)$$

Table 11 presents the top five recommendations generated for User 50.

Table 11: Top-5 movie recommendations for User 50 (Cluster 0)

Title	Genres	Avg. Rating	Count	Score
The Usual Suspects (1995)	Crime, Mystery, Thriller	4.239	46	16.321
Schindler’s List (1993)	Drama, War	4.070	50	16.003
American Beauty (1999)	Drama, Romance	4.154	39	15.323
One Flew Over the Cuckoo’s Nest (1975)	Drama	4.288	33	15.121
Seven (1995)	Mystery, Thriller	4.066	38	14.895

The example illustrates the implemented recommendation procedure. Ratings are drawn from other users in the assigned K-Means cluster, and eligible unseen movies are ranked by the stated score. The table reports the resulting suggestions and their within-cluster rating support.

No held-out ratings are used to assess these suggestions. The example therefore demonstrates the operation of the module, but does not establish that the suggested movies match the target user’s preferences or that cluster-based ranking is better than a global popularity-based baseline.

5.6 Discussion

The experiments provide evidence for descriptive differences between cluster-average profiles. In particular, the contrast between the Animation–Children–Fantasy pattern of Cluster 2 and the Mystery–Thriller pattern of Cluster 4 makes the final partition interpretable at the group level.

Interpretability and separation are different properties. The group averages can differ even when individual profiles overlap. The final Silhouette Score of 0.0638, together with the distribution of user-level values, limits how strongly the partition can be described as a set of distinct preference groups.

The Ward comparison gives context to the model choice. K-Means has a higher Silhouette Score and a larger smallest cluster at $K = 5$. These observations favour retaining the selected K-Means partition within the tested setting, but do not establish that it is the best method for user segmentation more generally.

For recommendation, the partition defines which users supply candidate ratings. This is a concrete use of the clustering output, but the benefit of restricting candidates to a cluster remains unmeasured. A held-out comparison with a global baseline is needed to determine whether segmentation improves the resulting suggestions.

5.7 Limitations

- **Weak cluster separation:** Both methods have low Silhouette Scores. The clusters should therefore be interpreted as approximate segments of the constructed profiles, rather than clearly separated user categories.
- **Differences within clusters:** A cluster-average profile does not fully describe each member. Individual users may differ from their group average in both the magnitude and direction of genre scores.
- **Missing entries and uneven rating support:** Unobserved user-genre entries are filled with zero, which does not distinguish missing evidence from an observed average equal to the user’s mean. Genre averages also have different numbers of supporting ratings, but this difference is not explicitly weighted in clustering.
- **Limited representation:** The features use genre labels and ratings only. Tags and temporal information are omitted. Movies with multiple genre labels contribute to several genre averages, so the features may share rating evidence. The effect of these design choices is not isolated experimentally.
- **Scope of the comparison:** Ward is evaluated only at $K = 5$ on the selected representation. The experiment does not optimize its cluster count separately or compare different linkage rules. The baseline comparison is also not a benchmark of current state-of-the-art methods.
- **Initialization assessment:** The multi-seed analysis measures variation in Silhouette Score and minimum cluster size. It does not measure agreement between user assignments across runs.
- **Unvalidated recommendation quality:** Recommendations are demonstrated without a held-out evaluation. No evidence is provided that the ranking rule or cluster-based candidate source improves recommendation quality over a baseline.
- **Approximate cold-start handling:** Favorite-genre inputs are converted into a manually defined profile, rather than a profile computed from observed ratings. The effectiveness of this mapping and the fallback rules has not been evaluated.

5.8 Future Work

The first priority is to evaluate recommendation quality on held-out ratings and compare the cluster-based module with a non-personalized baseline. User profiles, feature scaling, clustering, and movie-ranking statistics should be constructed from the training portion only. Relevant ranking metrics, such as Recall@ N and NDCG@ N , can then assess whether the suggested lists recover movies that users rated positively in the test portion.

Further clustering experiments should examine different cluster counts for both methods and measure agreement between K-Means assignments across initializations. This would distinguish stability of the partition from stability of aggregate metrics.

The representation should also be tested through controlled comparisons. Possible changes include distinguishing missing genre evidence from observed neutral scores, accounting for the number of supporting ratings, and incorporating tags or temporal information. These changes should be evaluated rather than assumed to improve clustering or recommendations.

Finally, the cold-start procedure could be examined using a defined simulation or evaluation protocol. This would help determine how well selected favorite genres approximate profiles derived from rating histories.

6 Conclusion

This project constructed a 19-dimensional user–genre representation from MovieLens ratings and applied K-Means to summarize 610 users into five preference segments. The selected cluster count reflects a practical choice based on the exploratory analysis of SSE, Silhouette Score, cluster sizes, and variation across seeds.

The final cluster-average profiles show interpretable differences, including contrasting patterns for Animation–Children–Fantasy and Mystery–Thriller. At the same cluster count and on the same standardized input, K-Means achieves a higher Silhouette Score than Agglomerative Ward, 0.0638 versus 0.0140.

These findings support retaining K-Means in the tested setting, while the low scores limit claims about separation of individual users. The recommendation module demonstrates how the partition can be used to generate movie suggestions, but its effectiveness has not been established. A held-out recommendation evaluation is needed to determine whether the segmentation provides a practical improvement over a global baseline.

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